

Contact

Email

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LinkedIn Profile:

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Address

Berlin

Expertise

- Organic Synthesis
- Perovskite Solar Cell Fabrication
- Computational Modelling
- Python & Data analysis

Languages

English: Full professional proficiency

German: A1 (Beginner)

Archana Sheshachala

Professional summary

Researcher with expertise in organic and materials chemistry, specializing in the design and synthesis of novel organic charge-transport materials, perovskite solar cell device fabrication, and optoelectronic applications. Experienced in interface engineering, charge-transfer studies, and device performance evaluation through integrated experimental and theoretical approaches.

Research experience:

Doctoral thesis

08/2020 - 06/2026

National Institute of Technology Karnataka (NITK), India

Thesis: Design, synthesis, and structural tailoring of heterocycle-based organic hole-transporting materials for perovskite photovoltaics.

- Developed dopant-free charge-transport materials to improve the performance and stability of perovskite solar cells.
- Studied structure–property–performance relationships.
- Performed theoretical and computational modelling of molecular and electronic properties.
- Developed and optimized perovskite solar cells, delivering strong photovoltaic performance with enhanced operational stability.
- Led collaborative research with teams at the University of Pennsylvania, University of Brunei, and Toin University of Yokohama
- Guided Master's thesis students in research projects.

Master's thesis

12/2018 - 06/2019

University of Mysore, Karnataka, India

- Synthesized and characterized quinazoline-based Schiff bases and their metal complexes, followed by evaluation of their antioxidant activity.

Bachelor's project

01/2017 - 05/2017

St Joseph's University, Bengaluru, India

- Synthesized carbohydrate-based metal-organic frameworks using a green chemistry approach.

ACADEMIC ACHIEVEMENTS

- ANRF International Travel (ITS) Grant Recipient
- PhD Fellowship funded by the Ministry of Education, Government of India

CONFERENCES ATTENDED

Poster Presentations

- International Conference on Chemistry for Sustainability (ICCS) 2025 – IIT Hyderabad
- Nature Conference: Advances in Perovskite Solar Cells 2025 – Stuttgart, Germany

REFERENCES

- Prof. Udaya Kumar D, National Institute of Technology, Karnataka, India
  udayakumar@nitk.edu.in
- Dr Kavya S. Keremane, University of Pennsylvania, Philadelphia, Pennsylvania, USA
  kxk5889@psu.edu

Education

Doctor of Philosophy in Chemistry 08/2020 - 06/2026
National Institute of Technology Karnataka (NITK), India

Masters in Chemistry 06/2017 - 06/2019
University of Mysore, Karnataka, India
Grade: 8.0 GPA

Bachelors in Science 06/2014 - 05/2017
(Physics, Chemistry, Mathematics)
St Joseph's University, Bengaluru, India
Grade: 82.94%

Skills

Experimental & Characterisation:	• Multistep organic synthesis • Chromatographic separation & purification techniques • Spectroscopic & analytical characterizations (NMR, UV–Vis, PL, FTIR, HRMS) • Perovskite solar cell fabrication and characterization
Computational Modelling & Simulation	• Quantum-chemical calculations and molecular dynamics simulations (Schrödinger Suite, Quantum ESPRESSO) • HPC cluster handling
Programming & Data Analysis	• Python for data analysis and predictive modeling

Publications

- **A. Sheshachala**, K. Keremane, et al., " A Molecular-Level Exploration of Dopant-Free Pyrazine-Derived Hole Transport Materials: Investigation of Interfacial Interaction in Perovskite Photovoltaics," 2025, **ChemPlusChem (Wiley)**, 90(9), e2500124.
- **A. Sheshachala**, K. Keremane, et al., "Strategic Molecular Design and Engineering of Star-Shaped Carbazole-Triphenylamine Hybrids for Dopant-Free Hole Transport Layer in Planar Perovskite Solar Cells," 2026, **Carbon Energy (Wiley) - accepted.**
- V. G. Bhat, K. Keremane, S. K. S., **A. Sheshachala** et al., "Dopant-free hydrophobic fluorene-based hole transport materials: impact of methoxy-substituted triphenylamine and carbazole peripheral groups on the performance of perovskite solar cells," 2025, **Sustainable Energy & Fuels (RSC)** 9(10), 2769–2781.
- V. G. Bhat, S. K. S., **A. Sheshachala** et al., "A 3,5-dicyano pyridine-core spiro-free hole-transport material for perovskite solar cells," 2026, **Chemical Communications (RSC)- accepted.**
- S. K. S., V. G. Bhat., Kavya S. K., **A. Sheshachala** et al., "Synergistic effects of linker and peripheral groups in carbazole-based dopant-free hole-transport materials for perovskite solar cells,"2026, **Materials Advances (RSC) -accepted.**